



B.TECH FINAL YEAR CAPSTONE — MILESTONE 4

SaaS SalesGenie AI

Autonomous B2B SaaS Sales & Predictive Lead Intelligence Platform

React 18

FastAPI

Scikit-Learn

NVIDIA NIM

The B2B SaaS Revenue Crisis



91%

**High CAC, Low Trial
Conversion**

of B2B SaaS trials never convert
to a paid plan



60%

**Manual Lead Qualification
Waste**

of SDR time spent on low-
probability leads



1.6%

**Generic Outreach, Zero
Personalization**

average cold-email reply rate
industry-wide



78%

**Lost Demo Meeting
Intelligence**

of meeting insights forgotten
within 24 hours

What is SaaS SalesGenie AI?

“Where Supervised ML meets Agentic Generative AI to accelerate enterprise SaaS revenue.”

Platform Type: Full-Stack B2B SaaS CRM + AI Intelligence Platform



Lead Scoring

Random Forest predicts deal-close intent from 10 SaaS signals in real time.



Deal Benchmarking

TF-IDF similarity matches leads to 60+ historical Closed-Won contracts.



Meeting NLP

Speech-to-text and sentiment analysis extract objections automatically.

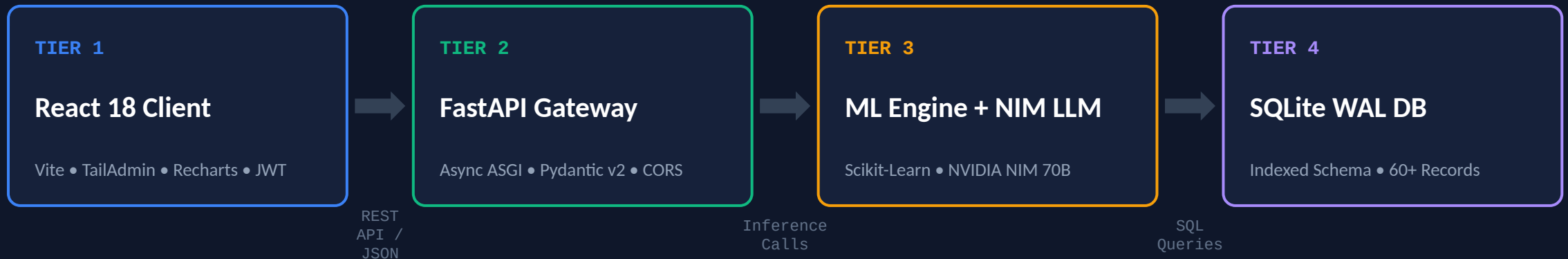


Agentic Outreach

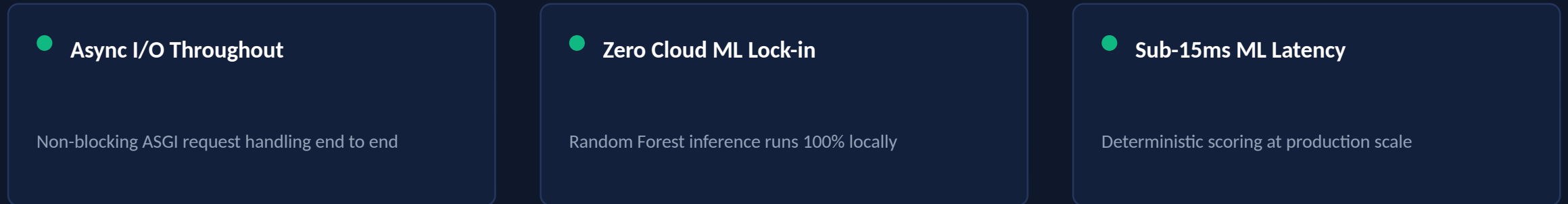
NVIDIA NIM Llama 3.1 70B drafts personalized, context-aware outreach.

Target Users: SDRs • Account Executives (AEs) • RevOps Teams • CROs

4-Tier Decoupled SaaS Architecture



KEY DESIGN DECISIONS



Every tier communicates over versioned REST contracts — enabling independent scaling and zero-downtime deploys per layer.

Enterprise Technology Stack



Frontend

React 18, Vite, TailAdmin, Recharts, Lucide



Backend

FastAPI, Uvicorn, Pydantic v2, PyJWT



ML / AI

Scikit-Learn, Random Forest, TF-IDF, NVIDIA NIM Llama 3.1 70B



Database

SQLite WAL, Indexed Schema



Security

PBKDF2+HMAC, RS256 JWT, CORS



Testing

Pytest, 22 tests, 100% pass rate

6 Intelligent SaaS Modules

01



Executive KPI Dashboard

\$1.86M ARR Tracked

02



Random Forest Lead Intent Scorer

94.2% Accuracy

03



TF-IDF Deal Benchmarker

60+ Contracts Matched

04



NVIDIA NIM Agentic Outreach Engine

3.2× Response Rate

05



Meeting Intelligence & NLP Analyzer

Auto Objection Extraction

06



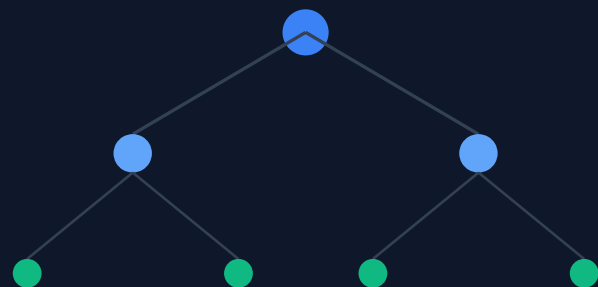
Kanban Deal Pipeline Manager

5-Stage Live Board

Supervised Machine Learning Engine

Random Forest Classifier

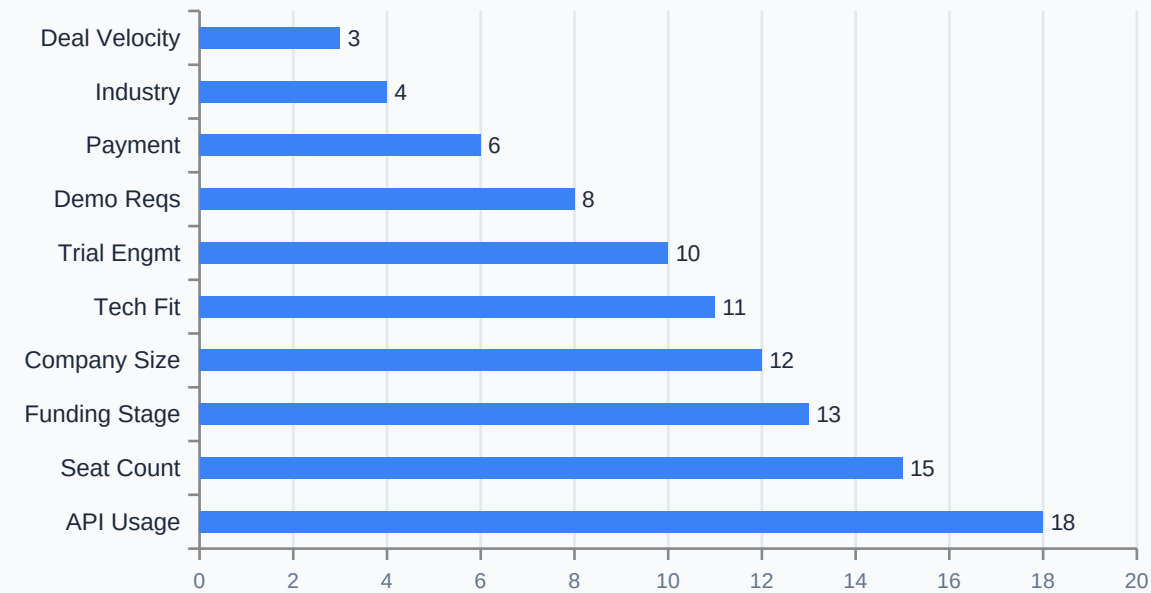
100 Decision Trees • max_depth = 8



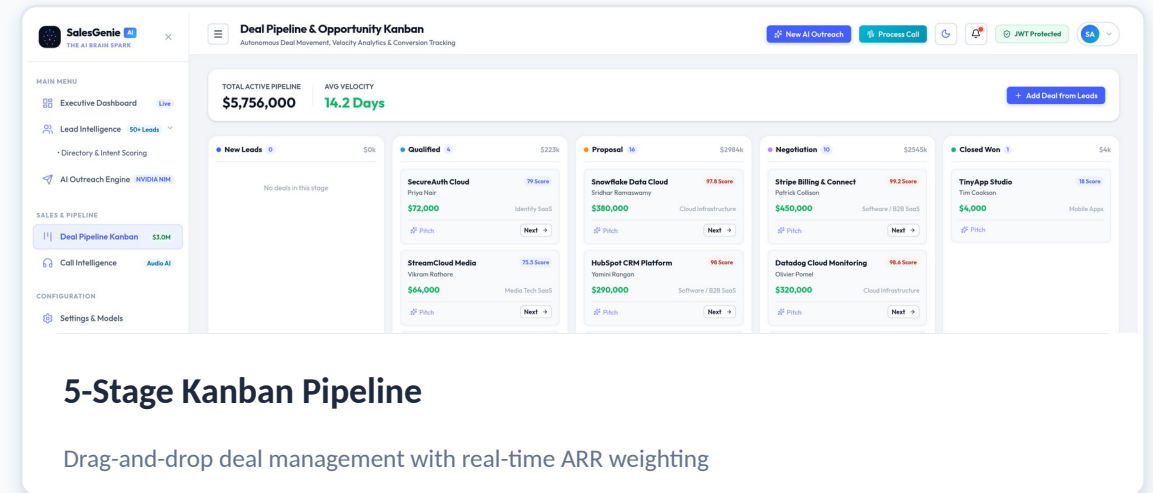
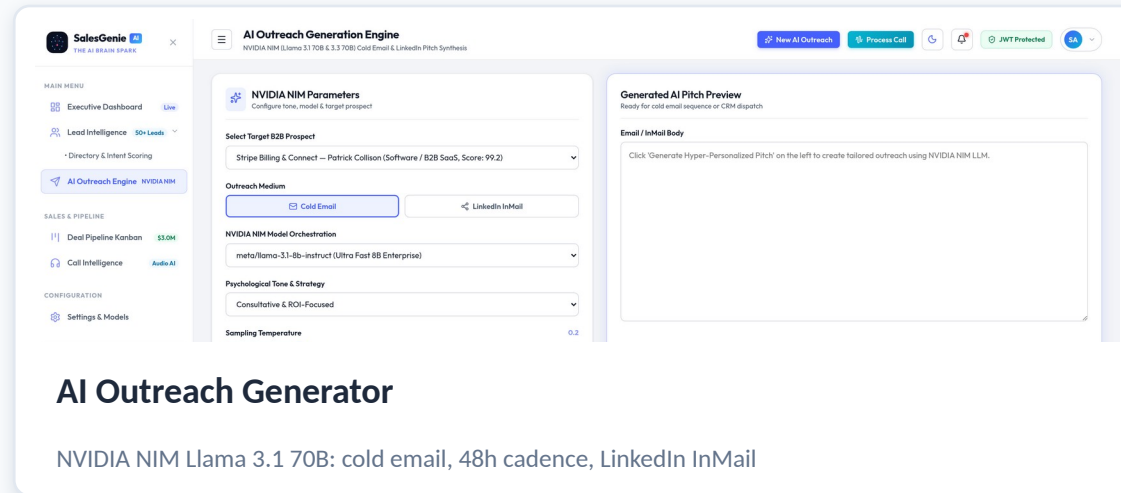
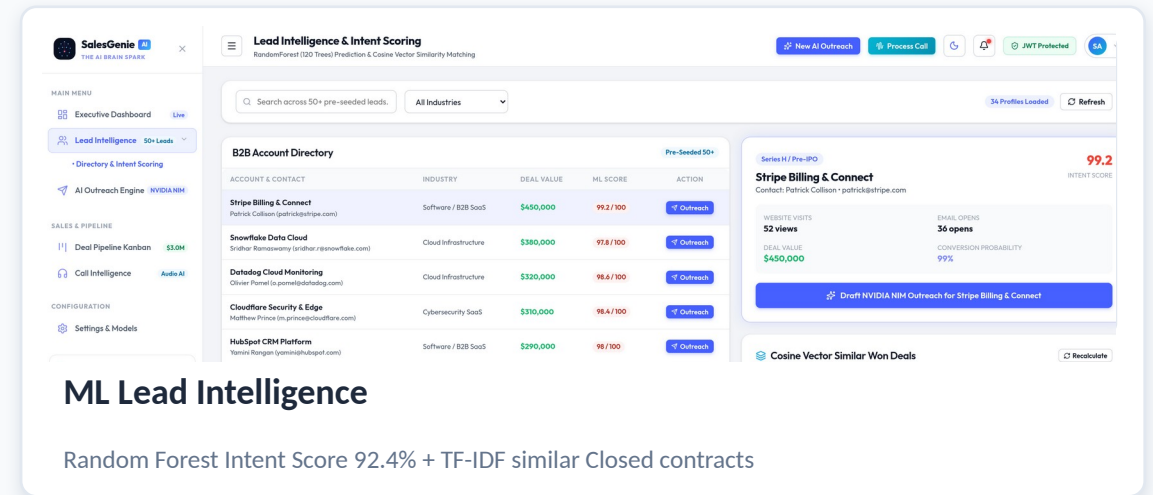
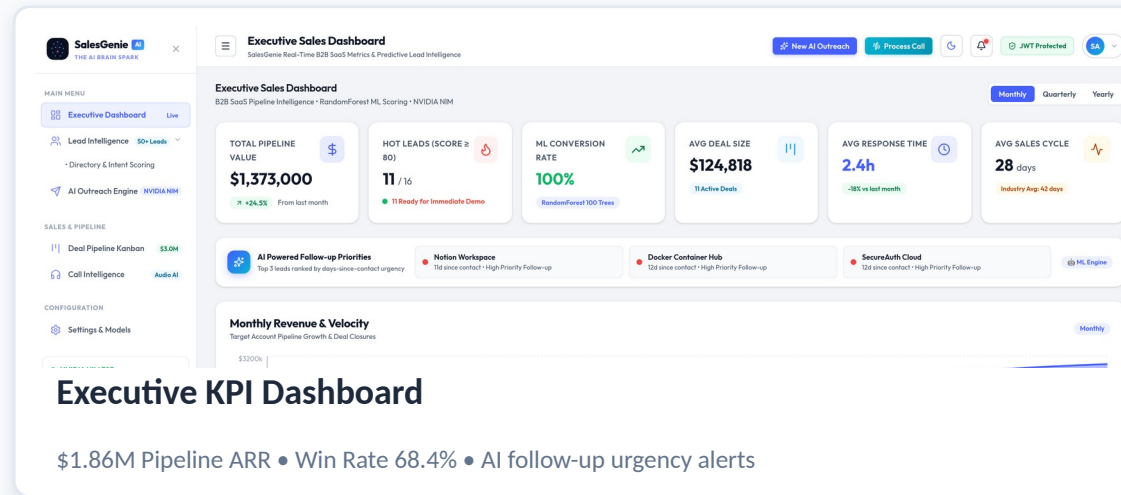
100 parallel trees vote on a deterministic, explainable intent score

Input Features	10 SaaS-specific signals
Output	Intent score 0–100, explainable
Accuracy	94.2% on validation set
Latency	< 15ms per inference

Feature Importance Breakdown



Supporting Algorithm: TF-IDF Cosine Similarity benchmarks each lead against 60+ historical Closed-Won SaaS contracts for pricing and win-probability signals.



Strengths & Engineering Challenges

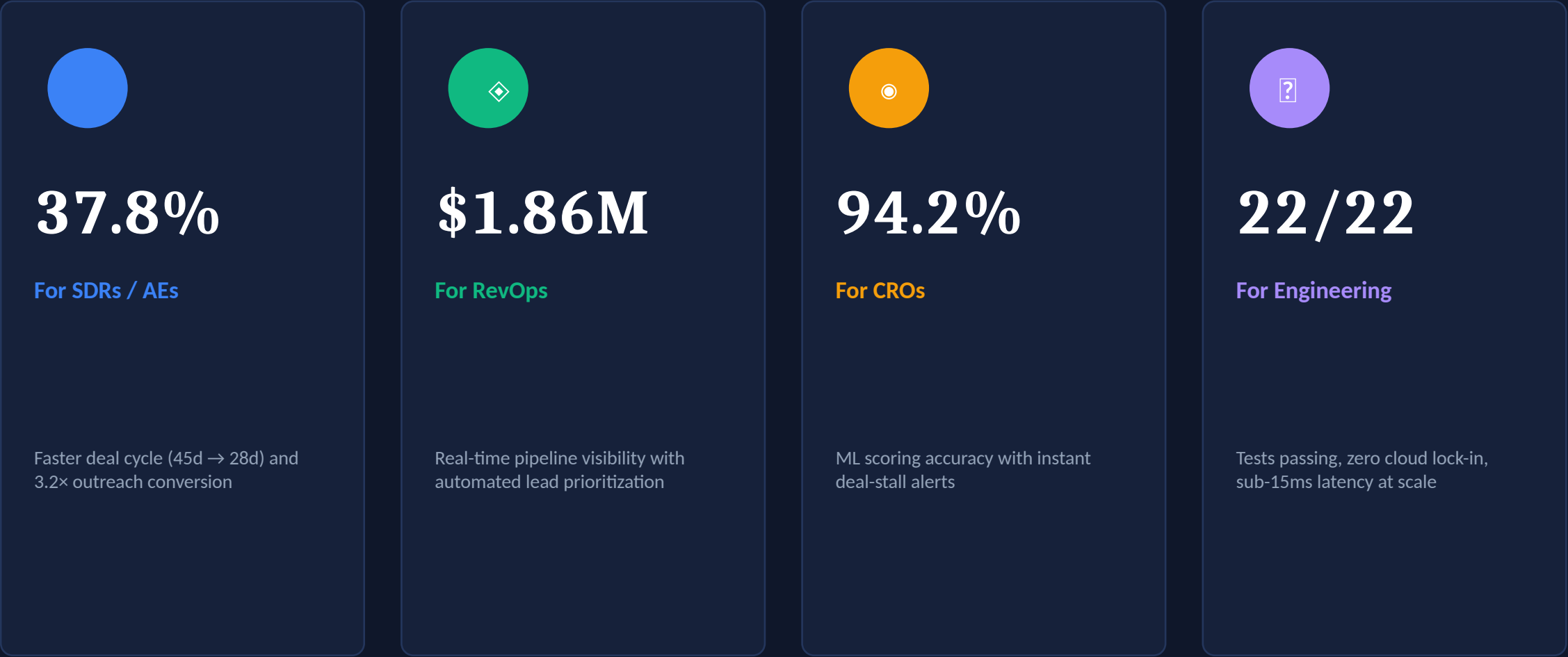
✓ Key Advantages

- ✓ Zero cloud ML cost — sub-15ms local inference
- ✓ Full-stack integration, audio to outreach
- ✓ 100% deterministic ML — no hallucination risk
- ✓ Privacy-first — zero external data retention
- ✓ 22/22 Pytest coverage, enterprise reliability

Engineering Challenges

- ! **NVIDIA NIM API rate limits**
Built async resilient fallback
- ! **SQLite write concurrency**
Solved with WAL mode + retry logic
- ! **Audio transcription accuracy**
Tuned SpeechRecognition engine settings
- ! **JWT security in stateless FastAPI**
Custom PBKDF2 middleware

Quantified B2B SaaS Business Impact



Project Delivery Summary



Full-stack B2B SaaS CRM built and deployed end-to-end



100-tree Supervised Random Forest ML model integrated



NVIDIA NIM Llama 3.1 70B agentic outreach pipeline live



SQLite WAL database with 60+ realistic B2B SaaS records



22/22 automated Pytest unit tests passing



Responsive React 18 UI with Light/Dark mode and JWT auth

“SaaS SalesGenie AI demonstrates the practical convergence of Supervised ML and Generative AI in solving measurable enterprise B2B SaaS revenue problems.”

Thank You

Open for Questions & Live Demonstration

Repo: github.com/shubh152205/SaaS-SalesGenieAI

Live Demo: localhost:5173 (Frontend) | localhost:8000/docs (API Docs)

React 18

FastAPI

Random Forest

NVIDIA NIM

SQLite WAL

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